

MoCHII User Guide

Abstract

How to build and run MoCHII (MOnte Carlo for H II regions): requirements, the complete input-parameter reference, output-file formats, worked examples keyed to the regression tests, and the runbook for adding an ion. The physics and validation are documented separately in docs/MoCHII_physics.pdf; the atomic-data fitting pipeline in docs/MoCHII_fitting.pdf.

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1. Requirements and build

- Intel oneAPI MPI Fortran (mpiifort/mpiifx; GNU mpif90 supported by the Makefile), HDF5 (default prefix /data/opt/hdf5_intel; override with make HDF5_PREFIX=...), CFITSIO.
- The SEDust dust-emission library, self-contained under SEDust/ in this tree. Build it once with `cd SEDust/sed && ./build_lib.sh` (→ SEDust/sed/lib/libsedust.a and its .mod files, which the Makefile links). The SEDust *data* directory is read at run time and resolved automatically relative to the executable (Section 2.10).
- Python 3 with numpy/scipy (h5py, astropy, matplotlib, PyNeb for the test and pipeline scripts).

Build:

```
cd MoCHII
(cd SEDust/sed && ./build_lib.sh)  # on a fresh checkout
make                               # -> MoCHII.x (always a full clean rebuild)
make DEBUG=1                       # bounds/traceback build
```

The SEDust library only needs the one-time `build_lib.sh` step; afterward `make` links the in-tree SEDust library. The Makefile intentionally performs a full rebuild every time (the MoCafe policy): there is no inter-module dependency tracking, and stale .mod files after editing `define.f90` otherwise cause runtime namelist errors.

Run:

```
mpirun -np 8 ./MoCHII.x input.in
```

MoCHII is MPI-only; the octree and all leaf arrays live in MPI-3 shared memory (one copy on each node).

2. Input parameters

All input is one `¶meters` namelist assigning fields of `par`. Parameters not listed keep the defaults shown. The switches compose: `gas_niter = 0` gives a single rates-only pass; `gas_niter > 0` iterates the ionization; `solve_te` adds the thermal balance; `use_metals` the trace-metal registry; `diffuse_field` the explicit diffuse field (forces case A); `refine_front` the I-front re-refinement; `ion_add_dust` the dust coupling.

2.1 Photons and RNG

parameter	default	meaning
no_photons	1e5	source packets in each iteration
no_print	1e7	progress-print stride
iseed	0	RNG seed (0 = from clock/pid)
launch_sequence	'random'	quasi-random launch: 'sobol' = Owen-scrambled Sobol coordinates for the stellar (and diffuse) packet launch, indexed by the global photon number (MPI-count-independent launch set); transport/scattering stay pseudo-random; prefer nphotons = 2^m . 'random' is the Mersenne Twister default
qmc_seed	12345	scramble key (launch_sequence='sobol'); fixed across the gas iterations; change it for independent replicates (their scatter is the RQMC error estimate)

2.2 Grid and dust density

parameter	default	meaning
grid_type	'car'	'amr' (octree) or 'car' (single-level Cartesian, leaf list in raster order; no tree and no geometry arrays in memory — cell centers and half-widths are computed analytically from the raster index; refine_front unavailable). 'uniform' is accepted as an alias for 'car'
car_walk	'dda'	car-grid traversal: 'dda' (default) = dedicated incremental Amanatides–Woo walks ($t_{\max}/t_{\Delta}$ per axis, comparisons and additions only — the fastest car walk); 'shared' = the octree walks with integer index arithmetic replacing the neighbor table, a verification mode that is bit-identical to 'amr'
amr_file	''	generic AMR leaf file (FITS/HDF5/text): columns x, y, z, level, nH [, metallicity, xHI, ndust]. If empty, a 'car' grid is built from the namelist instead ('amr' requires the file)
nx, ny, nz	1, 1, 11	'car'-from-namelist cell counts (cubic cells required: $2x_{\max}/n_x = 2y_{\max}/n_y = 2z_{\max}/n_z$)
xmax, ymax, zmax	1, 1, 1	'car'-from-namelist box half-extents [distance_unit]; box is $[-x_{\max}, x_{\max}]$ etc.
nH_const	-1	if ≥ 0 , override every leaf's n_H with this uniform value [cm^{-3}] (both 'car' and 'amr'); < 0 keeps the file/column density
density_file	''	read the leaf n_H from a uniform 3D density cube onto a 'car' grid (FITS 3D image, or the MoCHII HDF5 image format; $\text{NAXIS1}/2/3 = n_x/n_y/n_z$, values n_H [cm^{-3}]). $n_x/n_y/n_z$ come from the file, the box is xmax/ymax/zmax; an alternative to amr_file. reduce_factor > 1 down-samples; a rmax/rmin cut can mask the cube to a sphere
rmax, rmin	-999, 0	spherical density cut on leaf centers (both grids): $n_H = 0$ for $r > r_{\max}$ ($r_{\max} > 0$) and for $r < r_{\min}$ ($r_{\min} > 0$; a shell); radii in distance_unit
distance_unit	'kpc'	'kpc'/'pc'/'au'/'' (code units)

parameter	default	meaning
dust_model	'global_dgr'	leaf grey opacity: 'none', 'global_dgr' ($n_{\text{H}}c_{\text{ext}}\cdot\text{DGR}$), 'from_file' (ndust column), 'laursen09' (static x_{HI} column), 'laursen09_live' (computed x_{HI} , refreshed each iteration)
cext_dust	1.6059e-21	reference extinction per H [cm^2] at λ_{ref}
DGR, Z_global, Z_ref	1e-2, 0.0134, 0.0134	dust-model scalings
f_ion_dust	0.01	laursen09 survival of dust in ionized gas
f_pah, f_ion_pah	0, 0	PAH share of the reference extinction and its own ionized-gas survival (laursen09_live)

2.3 Ionizing/FUV band and the source

parameter	default	meaning
use_ion_band	F	must be T (the MoCHII transport)
nnu_ion	16	ionizing-band bins; gates use 32. With ion_align_edges on (the default) the bins are log-uniform within each threshold-bounded segment; off, they are a single log grid
eion_min, eion_max	13.598, 100	ionizing band edges [eV]
ion_align_edges	T	pin the ionizing-band bin edges onto the ionization thresholds (H I 13.6, He I 24.6, He II 54.4 eV plus the active metal edges) so no bin straddles one; segments between thresholds are log-uniform, and if the thresholds outnumber the bins the lowest-priority metal edges are dropped (H/He always kept), so it is safe at any nnu_ion. Removes the $\sim 18\%$ He-ionizing-photon excess of a coarse log grid crossing the 24.6 eV edge. Set F for the legacy single log grid
add_fuv	F	FUV extension: extra log bins on [efuv_min, eion_min] below the unchanged ionizing bins; transports FUV grain heating and FUV photoionization of low-threshold metals (Mg I, C I, Si, Fe I); luminosity stays the IONIZING luminosity (Q_H preserved, the FUV part rides on top)
efuv_min	6.0	FUV lower edge [eV]
nnu_fuv	8	FUV bins (16 recommended with metals)
tstar	-999	Planck source temperature [K]
ion_spectrum	"	global spectrum file (column 1 plus one value column); its column units are set by spectrum_type (§2.5)
spectrum_type	'shape'	column units of every spectrum-file slot (ion_spectrum, src_spectrum_file, ext_spectrum): 'shape' (arbitrary, normalized to the scale — legacy) or a physical type 'per_ev'/'per_hz'/'per_ang'/'per_um' (absolute); see §2.5
luminosity	unset	ionizing BAND luminosity [erg/s] of the (single) source; for a 'shape'/Planck source unset maps to 1.0 (legacy), for a physical-type file unset means DERIVE it from the file integral; with $n_{\text{source}} \geq 2$ it becomes the sum of src_lum (or the equal-split total when those are unset)
xs_point, ys_point, zs_point	0	single-source position (code units, recentered box; $n_{\text{source}}=1$)

Multiple internal point sources (the scalars above are the $n_{\text{source}}=1$ case):

parameter	default	meaning
nsource	1	number of internal point sources (0–16); 1 keeps the legacy scalars above, ≥ 2 uses the <code>src_*</code> arrays
src_x, src_y, src_z	0	positions of sources 1..nsource (code units)
src_lum	unset	ionizing luminosities [erg/s] of the sources; set all or none (none = equal split of luminosity); a partial set aborts
src_tstar	–999	Planck temperature [K] for each source (multi-source runs only; a single source uses <code>tstar</code>)
src_spectrum_file	”	one multi-column spectrum file for all sources: column 1 then one value column for each source (units set by <code>spectrum_type</code> , §2.5); each source is independently normalized so its ionizing segment carries <code>src_lum(i)</code> (a physical type with <code>src_lum</code> unset is derived from that column); overrides <code>src_tstar</code> ; too few columns abort, extra columns are ignored with a note

External isotropic field (composes with the internal sources; ON when `ext_intensity` > 0, an ISRF preset is named, or a physical-type `ext_spectrum` file is given):

parameter	default	meaning
ext_intensity	-1	band-integrated mean intensity J [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$] of the external field; the field is ON when > 0 and adds to the internal sources; entering ionizing power $L = \pi J A_{\text{surf}}$. With a physical-type ext_spectrum file or an ISRF preset it is optional: unset = the absolute J is DERIVED from the file/preset, set = the field is RESCALED to this band J (a note is printed)
ext_geometry	'rec'	illuminating surface: 'rec' (box faces, face-area-weighted — correct for a non-cubic box) or 'sph' (bounding sphere of radius rmax about the box center; needs $r_{\text{max}} > 0$)
ext_tstar	-999	external-field Planck temperature [K]
ext_spectrum	"	external-field spectrum: a 2-column file (units set by spectrum_type, §2.5; column 2 is an intensity density J) OR a reserved ISRF preset name 'draine'/'habing'/'mathis' (FUV-only, absolute, needs add_fuv); priority $\text{ext_spectrum} > \text{ext_tstar} >$ the global spectrum
ext_scale	1.0	dimensionless multiplier on an ISRF preset (ignored when $\text{ext_intensity} > 0$, which rescales the preset to a target band J)
source_geometry	'point'	legacy external-only shorthand: 'external', 'external_rec', or 'external_sph' forces nsource=0 with the external field on (needs $\text{ext_intensity} > 0$); 'point' keeps the composable form; 'slab' is the plane-parallel slab illumination below

2.4 Plane-parallel slab illumination

With source_geometry = 'slab' (which requires xy_periodic, §2.2) the box is an *xy-periodic slab*: the medium varies in z (uniform in x, y for a true 1D column with $n_x = n_y = 1$, or an arbitrary 3D tile with $n_x, n_y > 1$ repeated periodically), and light enters through the top ($+z$) and/or bottom ($-z$) face. Each face is either a collimated *beam* at incidence angle θ or an *isotropic* (Lambert) field, with its own strength. The single physical normalization is the flux F_z crossing the (horizontal) illuminated face: for a beam $F_z = \mu F_n$ ($\mu = \cos \theta$, F_n the flux normal to the beam), for the isotropic mode $F_z = \pi I$ (I the specific intensity). F_z sets the packet weight through $L = F_z A_{\text{zface}}$; every per-volume result is independent of the tile area. Point-observer peel-off is disabled under xy_periodic (the periodic images have no finite-distance observer); the emergent observable is the boundary intensity $I(\mu)$ (§3.3).

parameter	default	meaning
slab_faces	'top'	which z-face(s) are lit: 'top', 'bottom', or 'both'
slab_top_mode, slab_bot_mode	'beam'	per face: 'beam' (collimated) or 'isotropic' (Lambert)
slab_top_theta, slab_bot_theta	0	beam incidence angle from the inward normal [deg]; 0 is normal incidence
slab_top_phi, slab_bot_phi	0	beam azimuth [deg]
slab_top_source, slab_bot_source	0	per-face source strength (band-integrated): the specific intensity I [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$] for 'isotropic', or the beam flux F_n [$\text{erg s}^{-1} \text{cm}^{-2}$] for 'beam'
slab_nmu	20	number of μ -bins of the emergent $I(\mu)$ output

The spectrum is the global one (tstar / ion_spectrum). Both z-faces always escape (no reflecting boundary). launch_sequence = 'sobol' is supported (both modes). With dust scattering (ion_dust_scatter) the scattered/reflected component is included in the boundary $I(\mu)$.

2.5 Spectrum-file formats

A single parameter, spectrum_type (default 'shape'), fixes the column units of *every* spectrum-file slot (ion_spectrum, src_spectrum_file, ext_spectrum). The type sets the units; the slot sets the meaning: an internal-source file gives a luminosity density L_X , the external-field file gives an intensity density J_X . Column 1 and the value columns are:

spectrum_type	column 1	internal (cols 2+)	external (col 2)
'shape'	E [eV] $\uparrow$	arbitrary per eV (shape of L_E)	arbitrary per eV
'per_ev'	E [eV]	L_E [$\text{erg s}^{-1} \text{eV}^{-1}$]	J_E
'per_hz'	ν [Hz]	L_ν [$\text{erg s}^{-1} \text{Hz}^{-1}$]	J_ν
'per_ang'	λ [$\text{\AA}$]	L_λ [$\text{erg s}^{-1} \text{\AA}^{-1}$]	J_λ
'per_um'	λ [μm]	L_λ [$\text{erg s}^{-1} \mu\text{m}^{-1}$]	J_λ

The external intensity density J_X carries the same spectral unit as the internal L_X times an extra $\text{cm}^{-2} \text{sr}^{-1}$ (a mean intensity). All files are plain text with '#' comment lines; values outside the tabulated range are zero. src_spectrum_file is one multi-column file: column 1 shared by all sources, then one value column for each source (extra columns beyond nsource are ignored with a note, too few abort).

The abscissa is a sample grid, not bin centers. Column 1 lists the points at which the spectral density is *sampled*: the file describes a continuous function, defined between the points by linear interpolation, and no bin widths are attached to the rows. MoCHII integrates each of its own band bins on a 32-point sub-grid of the bin, interpolating the table at every sub-point (zero outside the tabulated range), so the file grid and the band grid are independent. The spacing is arbitrary — nonuniform grids are fine; place points densely where the spectrum varies rapidly, since only linear variation is captured between adjacent points. Abscissa values must be distinct (no duplicates); a 'shape' file must be ascending in E , while the physical types accept any row order (the table is sorted internally after the unit conversion, so a wavelength table may be given wavelength-ascending).

Absolute vs. shape normalization. A physical type ('per_ev'/'per_hz'/'per_ang'/'per_um') is *absolute*: the bin luminosities follow the tabulated integral directly. If the matching scale (`luminosity / src_lum(i) / ext_intensity`) is unset it is DERIVED from the file — each source from its column's ionizing-segment integral, the external field from its band-integrated J — and reported in the log. If a scale is set, the spectrum is RESCALED to it (a note is printed) and the file then acts only as a shape. With `add_fuv` an absolute spectrum fills the FUV bins from the same table, with no separate normalization. 'shape' keeps the legacy behavior: the shape is normalized to the scale in every case. A spectrum with no luminosity inside the band aborts, as does rescaling a file whose ionizing-segment integral is zero.

'shape' is per eV, not variable-agnostic. 'shape' reads its value columns as the shape of the per-eV density L_E on the E [eV] grid of column 1. A table of F_λ (or F_ν) values re-gridded onto E under 'shape' therefore omits the Jacobian and is wrong by a factor E^{-2} (the $d\lambda/dE$ conversion). To use a wavelength- or frequency-tabulated shape, give it with its matching physical type and set the scale explicitly — the type conversion then applies the Jacobian and the scale RESCALES the file to your target, so the file acts as a shape with the correct units. For example, a wavelength shape normalized to 10^{49} ionizing photons' worth of luminosity:

```
par%spectrum_type = 'per_ang'
par%ion_spectrum   = 'wave_shape.txt'
par%luminosity     = 3.0e38
```

A small 'per_ang' source file for two point sources (λ ascending; the second source is FUV-faint):

#	lambda[A]	L_lam,1	L_lam,2
	300.0	7.80e34	3.4e33
	500.0	6.05e34	1.2e33
	912.0	4.10e34	0.0

2.6 Initial state and iteration

parameter	default	meaning
xHI_init	1.0	initial $n_{\text{HI}}/n_{\text{H}}$ (an xHI file column wins); an ionized start (1e-3) converges much faster than neutral
xHeI_init, xHeII_init	1, 0	initial He fractions
He_abund	0.1	$n_{\text{He}}/n_{\text{H}}$
gas_niter	0	iterations (0 = a single rates-only pass)
gas_tol	1e-3	convergence tolerance on x_{HII} (measure set by conv_crit)
conv_crit	'cell'	convergence measure: 'cell' (cell maxima; stalls at the front-cell noise floor) or 'vol' (volume-integrated L_1 ; noise-robust, recommended). See §2.7
ion_relax	1.0	under-relaxation on x (< 1 damps front oscillation)
case_ab	'B'	case A/B recombination in the ionization balance, i.e. how the ionizing <i>continuum</i> is treated ('A' is forced by diffuse_field, which transports the ground-state recombination photons)
line_case	'B'	case of the SH95 H I/He II recombination- <i>line</i> tables. This is set by the optical depth of the Lyman <i>lines</i> , a different question from case_ab: a nebula whose diffuse continuum is transported still traps its Lyman lines and radiates case-B Balmer lines, so 'B' is the physical default for H II regions. 'A' selects the case-A tables (Lyman-line-thin gas); 'follow' ties the line case to case_ab
require_convergence	F	if the gas iteration has not converged at gas_niter, print a warning; when T, abort before writing output. The convergence status is recorded in the _rates header either way (CONVERGD, NITERDON, FINALDX, FINALDTE)

2.7 Convergence of the gas iteration

The gas iteration (gas_niter sweeps of transport $\rightarrow$ rate integrals $\rightarrow$ leaf solve $\rightarrow$ opacity refill) stops when the state stops changing between sweeps. Each sweep compares a *change measure* against a tolerance:

- ionization only (te_fixed, no thermal solve): converged when the x_{HII} change $<$ gas_tol;
- coupled (solve_te): converged when the x_{HII} change $<$ gas_tol *and* the T_e change $<$ gas_tol_te.

Two measures (conv_crit). Both are computed every sweep; conv_crit selects which one gates the loop.

'cell' the cell maxima $\max_{\text{leaf}} |\Delta x_{\text{HII}}|$ and $\max_{\text{leaf}} |\Delta T_e|/T_e$. Simple, but the ionization-front cells jitter with Monte Carlo noise and never settle, so $\max |\Delta x_{\text{HII}}|$ *stalls* at the front-cell noise floor ($\sim 2\text{--}4 \times 10^{-2}$ at 4×10^7 packets) and a smaller gas_tol is unreachable. This is the historical default (kept so the recorded gates reproduce).

'vol' the volume-integrated L_1 changes

$$\frac{\sum_{\text{leaf}} V |\Delta x_{\text{HII}}|}{\sum_{\text{leaf}} V x_{\text{HII}}} < \text{gas_tol}, \quad \frac{\sum_{\text{leaf}} V n_e |\Delta T_e|}{\sum_{\text{leaf}} V n_e T_e} < \text{gas_tol_te}.$$

Front jitter occupies little volume (a few thousand front cells against $\sim 10^5$ ionized ones), so it enters the norm suppressed by that ratio; the n_e weight drops the vacuum and `te_min`-pinned no-heating cells that otherwise dominate $\max |\Delta T_e|$ in FUV/PDR runs. These measures decay smoothly — the recommended production setting.

The Monte Carlo floor. Even the volume measures do not reach zero: they floor at a Monte Carlo level set by the packet count, scaling as $1/\sqrt{N_{\text{packets}}}$ ($\sim 3 \times 10^{-3}$ in x and $\sim 7 \times 10^{-3}$ in T_e at 8×10^6 packets). Reaching the floor *is* convergence — iterating further leaves the solution unchanged. So **set `gas_tol/gas_tol_te` just above the floor for the packet count in use**: a tolerance below the floor can never be met and the loop runs to `gas_niter` and warns. Rules of thumb (`conv_crit='vol'`): $\sim 5 \times 10^{-3}/10^{-2}$ at $\sim 10^7$ packets, loosened to $\sim 10^{-2}/2 \times 10^{-2}$ at $\sim 10^6$ (the reduced-count examples/ use the looser values for this reason).

When it does not converge. Hitting `gas_niter` without meeting the test is not fatal: the state is still written and a warning is printed on rank 0. Set `require_convergence = T` to abort before the output instead. Either way the `_rates` header records the outcome in `CONVERGD` (0/1), `NITERDON` (sweeps run), `FINALDX`, and `FINALDTE`. The derivation and the gate measurements are in `docs/MoCHII_physics.pdf`.

2.8 Thermal balance and metals

parameter	default	meaning
solve_te	F	bisection on T_e coupled with the ionization
te_fixed	1e4	fixed T_e [K] when solve_te is off
te_min, te_max	1e3, 5e4	bisection bracket [K]
gas_tol_te	1e-3	convergence on $\max \Delta T_e /T_e$
cooling_model	'local_ne'	line cooling in the thermal loop: 'local_ne' (default) suppresses each ion's cooling fit to the local n_e with a (T, n_e) table built from its n -level solve, so the thermal balance and the emergent lines are one physics; 'low_density' uses the $n_e \rightarrow 0$ fits directly (the coronal limit, exact in diffuse gas; reproduces earlier gates)
atomic_dir	'data/atomic'	coefficient files (relative to the run directory)
use_metals	F	registry cascade + metal line cooling
ci_model	'voronov'	collisional ionization: 'voronov' (default) or 'dere_hybrid' (Voronov $\times$ the Dere 2007/Voronov ratio; matters only in shielded low-ionization zones)
abund_C/N/O/Ne/S	HII20 values	$n(X)/n(H)$; an element is active when > 0
abund_Ar/Mg/Fe	0	further registry elements (gas-phase values; Mg/Fe are heavily depleted)
abund_Si/Cl/Ca	0	five-stage registry elements (gas-phase; Si and especially Ca heavily depleted, Cl nearly undepleted — Orion-like $3e-6 / 3e-7 / 2e-8$); adds [Si II] 34.8 μm , Si III/IV, [Cl II/III/IV] (5518/5538 density pair), Ca II, [Ca V]
ion_metal_abs	T	metal photoionization absorption in the band opacity (active with use_metals); the only gas opacity in the FUV bins

parameter	default	meaning
emis_output	F	write <code>_emis</code> (HDF5/FITS): leaf-by-leaf line emissivities + the state needed for maps (Section 3.)
hei_metastable	F	write <code>_heimeta</code> (HDF5/FITS): the He I 2^3S metastable density n_3 , its photoionization rate Φ_3 , plus n_H , n_e , T_e , and leaf geometry (the 10830 Å absorption diagnostic; Section 3.). With <code>add_fuv</code> and <code>efuv_min=4.78</code> the soft-UV part of Φ_3 is included
h_coll_effects	F	add the collisional H I Balmer component (excitation of neutral H) as separate hc rows/blocks; matters at fronts and hot cells
fe_levels_full	F	load the expanded Fe II/III models (<code>nlevel_fe_{2,3}_full.txt</code> , 16/34 levels) instead of the compact 13/14 for iron-line studies
diffuse_field	F	explicit ground-recombination packets
hei_diffuse	F	fourth diffuse channel (needs <code>diffuse_field</code>): He I excited-level recombination photons (19.82 eV, 584 Å, two-photon) — correct energy bookkeeping; moves Te by $\sim 1\%$ at HII40, He split unchanged
gaunt_vh14	F	free-free Gaunt factors from the van Hoof et al. (2014) table (<code>data/gauntff_vh14.dat</code>) instead of Hummer (1988); the two agree to $\lesssim 0.5\%$ in T_e
metal_ne	F	PDR: electrons from the metal cascade join the n_e closure (the only electrons beyond the I-front)
metal_heat	F	PDR: metal photoheating in the thermal balance
grain_pe	F	PDR thermal package: grain photoelectric heating + recombination cooling (Bakes & Tielens 1994) from the local FUV field (needs <code>add_fuv</code>), plus H-impact [C II] $158\ \mu\text{m}$ / [O I] $63\ \mu\text{m}$ cooling; scaled by <code>pe_scale</code> and the leaf dust amount
pe_scale	1.0	photoelectric-efficiency scale factor
use_sec_ion	F	secondary ionization by fast photoelectrons (Shull & van Steenberg 1985): a photoelectron with kinetic energy $E_0 = h\nu - E_{\text{th}}$ above 40 eV deposits only $f_{\text{heat}}(x)$ of its excess as heat and the balance drives secondary H I/He I ionizations (x = ionized fraction of the H+He nuclei); ~ 0 in fully ionized gas, matters at fronts / PDR / hard fields
metal_freefree	T	include the trace metals in the free-free cooling with the net ion charge $Z_{\text{ion}} = (\text{stage} - 1)$, not the nuclear Z ; a trace ($\sim 10^{-4}$ of the H/He free-free) correctness term, on by default

2.9 I-front re-refinement

parameter	default	meaning
refine_front	F	rebuild the octree on the I-front once
refine_iter	20	at which iteration
refine_lbase, refine_lmax	5, 7	base/front levels
refine_eps	1e-2	front flag: $\epsilon < x_{\text{HI}} < 1-\epsilon$
refine_dx	0.2	or a face-neighbor x_{HI} jump above this

2.10 Dust in the band and dust emission

parameter	default	meaning
ion_add_dust	F	dust absorption in the band (dusty Strömgren)
ion_dust_kext	"	kext table with EUV coverage (e.g. data/kext_albedo_WD_MW_3.1_60_D03.all_2003); required
ion_dust_scatter	F	sample dust scattering (HG, albedo weights)
dust_sed	F	SEDust stochastic dust SED at output time
dust_model_sed	'astrodust'	'astrodust'/'dl07'/'zubko'
sed_workdir	" (auto)	SEDust sed/ tree whose dielectric tables are read relative to it; blank auto-resolves to <executable dir>/SEDust/sed (the in-tree copy), so it normally needs no setting
sed_qtable, sed_sizedist	in-tree defaults	optics/size-distribution tables (relative to sed_workdir)
sed_NT, sed_Tlo, sed_Thi	200, 2.7, 5e3	SEDust temperature grid
dust_emis_bands	unset	up to 8 wavelengths [μm] (e.g. 8, 24, 70, 160): SEDust band emissivities of each leaf $\rightarrow$ _dustemis (blocks emis_dust/wl_dust [$\text{erg s}^{-1} \text{cm}^{-3} \mu\text{m}^{-1}$]); the Python map maker projects them into dust-band images (the IR is optically thin)
sed_pah_live	F	x_{HII} -dependent PAH survival in the emission: the leaf spectrum is assembled from the model channels with the PAH channel weighted by $(x_{\text{HI}} + f_{\text{ion,pah}} x_{\text{HII}})$ (needs a model with a 'PAH' channel, i.e. astrodust); turns the 8- μm map into the classic PAH ring at the ionization front

2.11 Peel-off imaging

parameter	default	meaning
ion_peel	F	after convergence, one extra transport pass peels direct (stellar + diffuse) and dust-scattered contributions toward the observers $\rightarrow$ <code>_image</code> (blocks <code>direct_ion/scatt_ion</code> , plus <code>_fuv</code> with <code>add_fuv</code>) [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$]
peel_bins	F	bin-resolved cubes <code>direct_cube/scatt_cube</code> (n_x, n_y, n_ν) in addition to the band-integrated channel images (hardness maps, synthetic EUV/FUV photometry); the third (bin) axis is labelled by the <code>E_bin/dE_bin</code> [eV] and <code>wl_bin</code> [$\text{\AA}$] arrays written once into the same file
save_direct0	F	also write the unattenuated direct image (<code>direct0_ion/_fuv</code> , and <code>direct0_cube</code> with <code>peel_bins</code>): the direct source with the path extinction $e^{-\tau}$ omitted, so <code>direct0/direct</code> = $e^{+\tau}$ is the line-of-sight optical depth toward the observer
obsx, obsy, obsz	unset	observer positions (or directions) in code units; or use <code>alpha/beta/gamma</code> angle lists [deg]; default: one observer on +z far away
distance	auto	observer distance (code units)
nxim, nyim	129, 129	image pixels
dxim, dyim	auto	pixel scale [deg]; auto-sized to cover the box

Peel-off also works with the external field: an external packet's direct peel uses the Lambert weight $\max(\hat{k}_{\text{obs}} \cdot \hat{n}_{\text{in}}, 0)/\pi$ about its entry-surface inward normal, so only the far-side surface contributes and the image is the background field seen through the medium (the cloud in silhouette); with `save_direct0`, `direct0/direct` = $e^{+\tau}$ is again the line-of-sight optical-depth map.

2.12 Output

`out_file` (base name; extension follows `file_format` = 'hdf5' or 'fits').

3. Output files

3.1 <base>_rates.h5

One dataset for each extension, leaf-ordered:

extension	content
Gamma_HI/HeI/HeII	photoionization rates [s^{-1}]
Heat_HI/HeI/HeII	photoheating per particle [erg/s]
Heat_dust	grain heating [$\text{erg s}^{-1} \text{cm}^{-3}$] (when ion_add_dust)
T_dust	equilibrium mixture dust temperature [K]
x_HI, x_HeI, x_HeII, n_e, T_e	converged state
x_<el>_stages	metal ion fractions (stage, leaf)
J_nu	band mean intensity (bin, leaf) [$\text{erg/s/cm}^2/\text{Hz/sr}$]
E_bin, dE_bin	band grid [eV]
LeafXYZ, LeafSize	leaf centers and full widths (code units; for slices/cutouts)

The file header also carries the convergence record of the gas iteration: CONVERGD (did it converge), NITERDON (iterations run), FINALDX (final max $|\Delta x_{\text{HII}}|$), and FINALDTE (final max $|\Delta T_e|/T_e$).

3.2 Line and emissivity outputs

<base>_lines.txt is a text table of integrated line luminosities on the converged state, one row for each line: element, ionization stage, wavelength [$\text{\AA}$], L [erg s^{-1}], and $L/L(\text{H}\beta)$. With emis_output the same line set is also written leaf-by-leaf to <base>_emis (HDF5, or FITS with file_format='fits') as emissivity blocks in $\text{erg s}^{-1} \text{cm}^{-3}$, where $\sum \text{emis} \cdot V = L_{\text{line}}$ and the block row order matches the text table. Every line comes from one of two sources: recombination-line tables (H and He, Table 1) and n -level statistical-equilibrium solves (the metals plus the optional collisional H I component, Table 2).

Table 1: Recombination-line blocks in <base>_emis; wavelengths in $\text{\AA}$ unless μm noted.

block	source	lines	weighting, present	when present
emis_h	Storey & Hummer (1995) H I, case set by line_case	H α 6565, H β 4863, H γ 4342, H δ 4103, Pa α 1.876 μm , Pa β 1.282 μm , Br γ 2.166 μm	always; $n_e n_{\text{HII}}$	
emis_heii	SH95 He II (case per line_case)	1640, 3204, 4543, 4686, 5413, 10126	only where an He III zone exists; $n_e n_{\text{HeIII}}$	
emis_hei	Porter et al. (2012/2013) He I, case-B collisional-radiative (density-dependent)	3889, 4026, 4471, 5876, 6678, 7065, 7281, 10830	where He II is present; $n_e n_{\text{HeII}}$	
emis_hc	25-level H I collisional solve	H α /H β /H γ collisional component (separate hc rows)	only h_coll_effects; n_{HI}	with

The He I Porter table is a case-B collisional-radiative total (density-dependent; it already includes collisional excitation from the 2^3S metastable), case B only; under case A a note is printed and the H I/He II tables switch to case A while He I stays case B.

The metal line blocks `emis_<el>_<stage>` follow the n -level solve. One block is written for every active registry ion (element abundance > 0) that has an `nlevel_<el>_<stage>` data file. Elements C, N, O, Ne, S are active by default; Ar, Mg, Fe, Si, Cl, Ca are activated by setting their `par%abund_<El>`. Only the principal lines are listed below — the full transition list of each block is in `<base>_lines.txt`.

Table 2: Metal collisional-line blocks (`emis_<el>_<stage>`); wavelengths in Å unless μm noted.

ion (block)	principal lines
c_2 [C II]	2325 (UV multiplet), 158 μm
c_3 [C III]	977, 1907, 1909, 178 μm
n_2 [N II]	3064/3071, 5756, 6550, 6585, 122 μm , 205 μm
n_3 [N III]	1750 (UV multiplet), 57 μm
o_1 [O I]	5579, 6302, 6366, 63 μm , 145 μm
o_2 [O II]	2471, 3727, 3730, 7321/7331
o_3 [O III]	1661/1666, 2322/2332, 4364, 4960, 5008, 52 μm , 88 μm
ne_2 [Ne II]	12.8 μm
ne_3 [Ne III]	3343, 3870, 3969, 15.55 μm , 36 μm
s_2 [S II]	4070/4078, 6718, 6733, 1.03 μm , 315 μm
s_3 [S III]	3722, 6314, 8831, 9070, 9532, 12 μm , 18.7 μm , 33.5 μm
<i>Optional — activate with <code>par%abund_<El></code></i>	
ar_3 [Ar III]	3006/3110, 5193, 7138, 7753, 8.99 μm , 21.8 μm
ar_4 [Ar IV]	2854/2869, 4713, 4741, 7170 (multiplet), 56 μm , 78 μm
mg_2 [Mg II]	2796, 2804
fe_2 [Fe II]	1.26/1.32 μm , 5.34 μm , 17.9 μm , 26 μm (40 lines total)
fe_3 [Fe III]	4659, 4703, 4881, 5011, 22.9 μm (38 lines; 16/34-level with <code>fe_levels_full</code>)
si_2 [Si II]	2335 (multiplet), 34.8 μm
si_3 [Si III]	1206, 1883, 1892, 38 μm
si_4 [Si IV]	1394, 1403
cl_2 [Cl II]	3587/3679, 6164, 8581, 9126, 14.4 μm , 33 μm
cl_3 [Cl III]	3344/3354, 5519, 5539, 8480 (multiplet)
cl_4 [Cl IV]	3120/3205, 5325, 7263, 7533, 11.8 μm , 20.3 μm
ca_2 [Ca II]	3935, 3970, 7293/7326, 8500/8544/8665
ca_5 [Ca V]	3999, 5311, 6088, 3.05 μm , 4.16 μm , 11.5 μm

Some registry ions are tracked in the ionization balance but carry no line block because CHIANTI v11 has no level/collision data for them: Ar II, Cl V, Ca III, Ca IV.

State blocks. Beyond the emissivities, <base>_emis carries the state a map needs without the rates file: LeafXYZ (leaf centers) with the DIST_CM keyword, LeafSize (leaf edge length; $V = (\text{LeafSize} - \text{DIST_CM})^3$), n_H, n_e, T_e, x_HI, x_HeI, x_HeII, and the x_<el>_stages metal-fraction blocks (same layout as the rates file). The Python EmisData reader (tools/python/mochii_output.py) turns these blocks into line maps without the rates file.

3.3 Spectral and image outputs

file	content
_nebcont.txt	nebular continuum L_ν (fb+ff tables, Milne, two-photon)
_dustir.txt	equilibrium-mixture dust IR spectrum (1–1000 μm) with the energy-closure header
_dustsed.txt	SEDust stochastic dust SED with PAH bands (when dust_sed)
_dustemis.h5	(with dust_emis_bands) SEDust band emissivities of each leaf, same block layout as _emis — readable by EmisData for dust-band maps
_heimeta.h5	(with hei_metastable) the He I 2^3S diagnostic: n_2s3 [cm^{-3}] (with the DIST_CM and A31 keywords), Phi_3 [s^{-1}], n_H, n_e, T_e, LeafXYZ, LeafSize — read by the Python HeiMetaData class for column and 10830 Å optical-depth maps, and convertible to a LaRT scattering-medium input by heimeta_to_lart.py
_image.h5	(with ion_peel) peel-off images for each observer: direct_ion/scatt_ion (+_fuv) [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$], suffix _obs<k> when several observers; with peel_bins, direct_cube/scatt_cube (n_x, n_y, n_ν) and the E_bin/dE_bin/wl_bin bin-grid arrays; with save_direct0, the unattenuated direc0_ion/_fuv (or direc0_cube)
_slab_Imu.txt	(with source_geometry = 'slab') the emergent intensity $I(\mu)$ at the two z-boundaries: columns μ , I_{top} , I_{bot} [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$] on slab_nmu bins ($F = 2\pi \int I \mu d\mu$), with the reflected / transmitted / absorbed energy budget (fractions of the entering L) in the header

3.4 Reading outputs in Python

tools/python/mochii_output.py is an importable module (works directly in a notebook) that reads every section file (HDF5 or FITS; read_sections), the line table (read_lines_table), and the text spectra (read_spectrum), and builds 2D maps from the emissivity file:

```
import sys; sys.path.insert(0, ".../MoCHII/tools/python")
import matplotlib.pyplot as plt
from mochii_output import EmisData

d = EmisData("run_emis.h5")
d.info()                                # blocks, fields, line count
```

```

d.line_list()                                # every stored line (block, wl)

# ready-made figures (notebook): returns (fig, ax)
fig, ax = d.plot_line("o_3", 5007.)          # log S map
fig, ax = d.plot_line("h", 6563., unit="intensity",
                      photons=True)         # photons/s/cm^2/sr
fig, ax = d.plot_field("T_e", weight="EM")
fig, axs = plt.subplots(1, 2); d.plot_line("s_2", 6717., ax=axs[0]) \
    ; d.plot_line("s_2", 6731., ax=axs[1])   # doublet side by side

# raw arrays, when the figure is yours to build
img, ext = d.line_map("o_3", 5007., axis="z", npix=512)
img, ext = d.line_map("h", 6563., unit="intensity", photons=True)
img, ext = d.field_map("T_e", weight="EM")   # LOS-weighted average
Q = d.line_luminosity("h", 4861., photons=True) # photon rate [1/s]

```

Line maps are flux-conserving: each leaf's luminosity is deposited with exact area overlap onto the pixel grid (AMR leaf sizes handled), so $\sum SA_{\text{pix}} = L_{\text{line}}$ to machine precision. `unit='flux'` (default) gives surface brightness [$\text{erg s}^{-1} \text{cm}^{-2}$]; `unit='intensity'` gives specific intensity [$\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$] = $S/4\pi$ (isotropic emission, optically thin); `photons=True` converts either to photon units (divides by hc/λ of the line). A `slab=(lo,hi)` cut restricts the line of sight. Field maps average `T_e`, `n_e`, `x_HI`, ... along the line of sight with weight 'EM' ($n_e^2 V$), 'ne', 'nH', or 'V'. The same module doubles as a command-line tool for quick looks:

```

python3 mochii_output.py run_emis.h5          # list lines
python3 mochii_output.py run_emis.h5 --line o_3 5007 \
    --npix 512 --log --out o3.png
python3 mochii_output.py run_emis.h5 --line h 6563 \
    --unit intensity --photons                # photons/s/cm^2/sr
python3 mochii_output.py run_emis.h5 --field T_e --weight EM

```

3.5 Slices and cutouts

A *slice* is the actual leaf value on a plane (the cell that contains each pixel), distinct from the line-of-sight *projection* above — a slice resolves the ionization front sharply where a projection blurs it. `tools/python/leaf_field.py` holds the backend (`leaf_slice`, the nearest-leaf fallback `leaf_slice_nn`, and `plot_slice`); `mochii_output.py` exposes it on the output files:

```

from mochii_output import EmisData, RatesData
r = RatesData("run_rates.h5")          # leaf state
fig, ax = r.plot_field_slice("T_e", axis="z", coord=0.0)
fig, ax = r.plot_field_slice("x_HI", axis="z", coord=0.0, log=True)

```

```
img, ext = r.slice_field("n_e", axis="x", coord=0.0,
                        bounds=(-2,2,-2,2)) # cutout window
```

```
d = EmisData("run_emis.h5")
fig, ax = d.plot_line_slice("o_3", 5007., axis="z") # emissivity slice
```

Exact slices need the leaf full widths: the `_emis.h5` file always carries `LeafSize`, and `_rates.h5` does too; a rates file written before `LeafSize` existed falls back to a nearest-leaf slice (a one-time note is printed). The command line takes `--slice FIELD --axis z --coord 0` on either file.

3.6 He I 2³S metastable and 10830 Å absorption

The `_heimeta` file (`hei_metastable`) is read by the `HeiMetaData` class, which adds a line-of-sight column map and the 10830 Å line-center optical depth:

```
from mochii_output import HeiMetaData
d = HeiMetaData("run_heimeta.h5")
d.info() # n_2s3, Phi_3, line constants
img, ext = d.column_map("n_2s3") # LOS column [cm^-2]
img, ext = d.tau10830_map(v_turb=0.0, # tau_0 map (km/s turbulence)
                        component="doublet") # blended J'=1,2 pair
fig, ax = d.plot_column("n_2s3", log=True)
fig, ax = d.plot_tau10830(component="all")
```

`column_map` is the same flux-conserving deposit as the line maps with weight nV , so $\sum NA_{\text{pix}} = \sum nV$ to machine precision. `tau10830_map` deposits the line-center opacity $\kappa_0 = \frac{\sqrt{\pi} e^2}{m_e c} f \frac{\lambda}{b} n_3$, $b = \sqrt{2kT_e/m_{\text{He}} + v_{\text{turb}}^2}$, so a pixel value is $\int \kappa_0 dl$. `component` selects the resolved $J' = 0$ line (10832.06 Å), the blended $J' = 1,2$ 'doublet' (10833.22/10833.31 Å, 0.089 Å apart, inside the thermal Doppler width), or 'all'; the constants (λ , f , A) come from `data/atomic/hei_metastable.txt`. The map warns when line-center $\tau_0 > 1$: the diagnostic assumes optically thin 10830 (line transfer is out of scope). The command line takes `--column n_2s3` or `--tau10830 --component doublet --vturb 5`.

Where line transfer *is* wanted (resonant-scattering emission profiles from the metastable medium), `heimeta_to_lart.py` (in `tools/python/`) converts a `_heimeta` file into a scattering-medium input for the resonance-line code LaRT (line HeI_10833): `--format car` (default, uniform grids) writes the all-in-one Cartesian file with `gasDen = n3`, `T = Te`, and zero velocities; `--format amr` writes the generic AMR table with `n3` in the `n_ion` column. `--field` selects a different leaf field for the scatterer. A worked end-to-end case (converted grid + LaRT namelist) is in `examples/hei_metastable/lart/`.

3.7 Building AMR grids

`tools/python/AMR_grid/` builds the generic AMR octree file the code reads (`grid_type='amr'`): `AMR_grid.py` (the `AMRGrid` class — uniform, density-gradient, or resolution refinement; `set_density/set_neutral_fraction`; write to

HDF5/FITS/.fits.gz/text with columns `x,y,z,level,nH[,metallicity,xHI,ndust]`) and `make_amr_sphere.py` (a sphere/shell CLI). AMRGrid carries the same `slice/plot_slice/cutout` so a grid can be inspected before a run.

4. Worked examples (the regression tests)

Each tests/ directory is self-contained: grid builder or a symlinked grid, the input file, and a `check_*.py` gate script.

directory	what it demonstrates / gate
<code>g0_gamma</code>	rate integrals vs. analytic attenuation (bias +0.004%)
<code>g1_stromgren</code>	Strömgren sphere, case B fixed T_e ; AMR $\equiv$ uniform
<code>g2a_thermal</code>	thermal H/He nebula vs. the 1D exact reference
<code>g2_hii</code>	MOCASSIN Lexington HII20/HII40 benchmarks + line fluxes
<code>g4_refine</code>	I-front re-refinement vs. native level 7
<code>g4_tng</code>	Illustris-TNG post-processing demo (shadows, maps)
<code>d_dusty</code>	dusty Strömgren; scattering bracket; T_d /IR
<code>peel</code>	peel-off images vs. the optically thin analytic flux (+0.017%); bin cubes
<code>uni_dda</code>	uniform grid: shared walk $\equiv$ octree (bit-identical); DDA walk to rounding
<code>pdr</code>	FUV/PDR switches: carbon-electron shell, photoelectric T_e
<code>ext_field</code>	isotropic external field: interior $J = J_{\text{ext}}$ to 0.2% (rec and sph)
<code>multi_src</code>	multiple point sources: superposition, spectrum resolution, one-file spectra
<code>peel_ext</code>	external-field peel: silhouette flux vs. JA_{proj}/d^2 , chord τ spectrum
<code>qmc</code>	quasi-random launch: Sobol generator vs. <code>scipy</code> + Owen-net balance; the fixed-state variance gate (cell-wise Γ_{HI} gain $29\times$ at 2^{17} , $222\times$ at 2^{20})
<code>hei_meta</code>	He I 2^3S metastable: OH18 analytic limits, module n_3 vs. the Python closed form, Φ_3 from the band tally, and the 10830 Å column/optical-depth diagnostic (conservation + hand integral)

A minimal thermal + metals + dust input (from `d_dusty`):

¶meters

```

par%no_photons = 4.0e7           par%iseed = 1234
par%grid_type = 'amr'           par%amr_file = 'amr_L6.fits'
par%distance_unit = 'pc'        par%dust_model = 'global_dgr'
par%use_ion_band = .true.       par%nnu_ion = 32
par%eion_min = 13.598           par%eion_max = 100.0
par%tstar = 4.0e4               par%luminosity = 3.178e38
par%xHI_init = 1.0e-3           par%He_abund = 0.1
par%gas_niter = 60              par%gas_tol = 2.0e-3
par%solve_te = .true.          par%atomic_dir = '../data/atomic'
par%use_metals = .true.
par%ion_add_dust = .true.

```

```

par%ion_dust_kext = '../data/kext_albedo_WD_MW_3.1_60_D03.all_2003'
par%cext_dust = 4.868e-22      par%DGR = 1.0
par%out_file = 'run.h5'      par%file_format = 'hdf5'
/

```

5. Adding an ion

Adding an ion touches only data:

1. `tools/fitting/fit_cooling.py`: add the ion with a T_i ladder guessed from its `.elvlc` level energies; run; check the reported max error and the overlay figure.
2. `tools/fitting/fit_nlevel.py`: add the ion with the level count needed for its diagnostics; run.
3. `tools/fitting/make_element_data.py`: add the element to `ELEMENTS` (Z , stage count) and its thresholds; charge exchange entries where sources exist; run. The CHIANTI `.rrparams/.drparams` types 1–3 are handled (RR/RR2/DR/DR2 lines).
4. Fortran: a `par%abund_<El>` field and one entry in the `species_setup` name/abundance lists (only for a NEW element; new stages of a known element need nothing).
5. Verify: extend `verify_nlevel_pyneb.py` with a diagnostic ratio where PyNeb has data; run a smoke test and check the new lines in `_lines.txt`.

Data caveats met so far: Ar II has no CHIANTI v11 level/collision data (stage tracked without lines); Fe III collision strengths differ from PyNeb's BB14 by up to $\sim 16\%$ in 4658/4702 (known spread for iron); charge-exchange stage labels differ between the Huang (2023) and MOCASSIN transcriptions for Mg/Fe (the Huang labels are adopted; see `make_element_data.py`).

6. Practical notes

- **Bins**: 32 bins give $\sim 2\%$ rate-integral discretization; use 64 for sub-percent work. The `add_fuv` extension carries its own bins below `eion_min`, so the ionizing resolution is unaffected.
- **FUV and the PDR**: without `add_fuv` there is no radiation beyond the I-front and the metals there recombine toward neutral; with it, FUV penetrates the I-front and photoionizes the metal stages whose thresholds lie below 13.6 eV (C I 11.3, S I 10.4, Si I 8.2, Fe I 7.9, Mg I 7.6 eV, ...), producing the PDR-side C II and Mg II. In the FUV band the gas opacity is exactly this low-threshold metal photoionization (`ion_metal_abs`, on by default; H and He do not absorb below 13.6 eV), so the FUV is attenuated by that photoionization together with dust — only where those metals are already ionized does dust extinction alone remain. The metal stage fractions iterate with this opacity self-consistently; the corresponding metal photoheating and the grain photoelectric heating are not in the thermal balance, so PDR-zone temperatures are indicative only.

- **Convergence:** use `conv_crit = 'vol'` for production — the cell-max criterion stalls at the front-cell noise floor (and, with `add_fuv`, at no-heating cells where T_e pins to `te_min`). Both measures are printed each iteration. The explicit diffuse field smooths the front and converges far better.
- **Starts:** prefer an ionized initial state; a neutral start advances the front $\sim$ one cell per iteration.
- **text tables:** must cover the band (the astro dust table stops at 912 Å; use the D03 table for EUV work). Tables are sorted on read (the D03 file has an out-of-order seam).
- **Re-refinement:** one event per run; the old shared-memory windows are reclaimed only at exit.
- **SEDust:** `sed_workdir` must point at a SEDust `sed/` tree; the exact stochastic solve costs $\sim 0.05\text{--}0.1$ s for each gas leaf (distributed over ranks).